Evidence files: Assel Saginayeva, AIB-2602 · BA2026 Assignment 3

Part 1: Week 6 marketing comparison. Part 2: Week 7 AI/data workflow and risk register.

Part 1. Week 6 artifact: marketing comparison

WEEK 6. MARKETING COMPARISON

Student: Assel Saginayeva Group: AIB-2602 Company B: JSC Bayan Sulu (Company A: Ferrero)
Goal: compare the SAME marketing task performed by a human-only approach and an AI-supported approach. Use documented facts where available; clearly label any hypothetical workflow or predicted KPI.

A. Customer and marketing context

Customer segment (who, needs, evidence/source):
Who: halal-seeking families in the Gulf (UAE first) who buy sweets to share at home, with guests and on holidays – a possible new market for Bayan Sulu, not yet a current one [4].

Needs: a clear halal mark and readable label, good taste, a fair price and a brand they can trust.
Evidence: at Gulfood, Dubai (26–30 Jan 2026), its first international exhibition, Bayan Sulu held B2B meetings with UAE partners and noted strong demand for halal-certified products [4]; it already exports about 40% of output to 14 countries [2].

Product/service and value proposition:
Product: Bayan Sulu packaged confectionery – chocolate, candies, caramel, waffles, cookies, marshmallow and marmalade (300+ products), made in Kostanay since 1974 [2].

Value proposition: "sweets from a brand families have trusted for 50+ years" [4] – clearly labelled, halal-marked confectionery for family sharing at a fair price, as an alternative to global brands such as Ferrero.

Customer journey: Awareness → Consideration → Purchase → Experience → Retention.
Identify one friction point:
Awareness: Gulfood stand and B2B meetings with UAE partners [4] → **Consideration:** "Is it halal? Is it good?" → **Purchase:** one importer → supermarkets and online shops → **Experience:** "same taste every time" → **Retention:** loyalty offer after the first purchase, only with consent [5].

Friction point (Consideration): Gulf families do not know the brand yet. Without a clear halal mark, Arabic/English information and online reviews, they choose a familiar global brand instead.

So the launch message must answer "Is it halal? Is it good?" in the customer's own language before any price or loyalty offer can work. (Reasoned from [4]; not measured.)

B. Same task, two approaches

Specific marketing task (e.g., draft a customer email; select audience; analyse feedback):
Promotion: draft the launch ad messages for halal-seeking Gulf families (Arabic, English, Kazakh) that answer "Is it halal? Is it good?" – hypothetical workflow, one product line.

Criterion	Human-only	AI-supported
Inputs / data	Gulfood notes and partner meetings [4], shop visits, distributor advice, brand book.	Same facts plus online reviews and UAE competitor ads/prices in Arabic and English, given to a generative AI tool. No personal

		data.
Steps & responsible actor	Marketing manager writes the brief → one agency writes one campaign → translator translates → brand manager approves.	Marketer writes brief and prompt → AI drafts many versions per group and language → native speaker and brand manager edit and choose.
Time / resources	Hypothetical: weeks of agency and translator work; one message for everybody.	Hypothetical: drafts in hours; costs are the AI tool and reviewer time. Volume rises, review work rises too.
Expected quality	Strong brand voice and local feel, but few versions and slow to change.	Many tailored versions, but risk of off-brand tone, bad translation or invented claims.
Privacy / bias risks	Low privacy risk; one team's view of a new market may miss local taste.	No personal data in prompts; AI may miss local taste or use stereotypes; uneven quality across languages [7].
Human review & approval	Brand manager signs off the final campaign.	Mandatory: native speaker + brand manager approve every ad; quality team checks halal/label claims; approval saved with name and date. AI never publishes [5], [7].

Use the rows above for: inputs/data; steps and responsible actor; time/resources; expected quality; privacy/bias risks; human review and approval.

C. Method and result

Comparison method: what input, prompt/instructions, sample or evaluation criteria did you use?
Method: one real case (Gulfood 2026 [4]) run through the 4Ps and customer journey [1]; for each item the same task was described twice – people only vs. with AI (photo essay, Visual 5). Same input for both: brand facts from [2], [4], the target segment above, and the halal/label rules.

AI instruction (illustrative): "Write 5 short ad messages for Gulf families about Bayan Sulu sweets in Arabic, English and Kazakh. Use only these facts; no health claims; mention the halal mark only if certified." **Criteria:** speed, number of versions, translation quality, brand voice, cultural fit and compliance – each row ends with a named person who checks.

Observed result or reasoned expectation (do not invent measured figures):
Reasoned expectation (no measured figures): AI does better at **speed** – many ad versions, one per group, in three languages, plus fast scanning of online reviews and competitor prices.

People do better at trust – local taste, partner meetings, halal and label rules, and the voice of a 50+ year brand [4]. Finding: AI prepares the options, a named person chooses (photo essay, Visual 6).

Which approach better serves the customer? What trade-off remains?
AI-supported with human sign-off better serves Gulf families: they get clearer messages in their own language sooner, and every claim is checked before it reaches them.

Remaining trade-off: speed and volume vs. accuracy and brand trust – a bad translation or a false halal/health claim could damage a brand the market does not know yet, so a native speaker and the brand manager must approve every ad [7].

D. Company A → Company B transfer

Documented comparator practice + source; verdict (ADOPT / ADAPT / PILOT / DELAY / REJECT) and why:
ADOPT – AI review and price scanning (industry practice): cheap and low risk; useful research before entering the Gulf. **ADAPT – Ferrero customer groups in Google BigQuery** ("Raw Data" by fifty-five, segments consumers for personalised content) [8]: Bayan Sulu has much less online data, so start with simple groups from distributor sales, with consent [5].

PILOT – Ferrero USA AI demand forecasting with Alloy.ai [10]: one product line, one UAE channel; return to manual planning if forecast error > 25% for 2 months. **DELAY – Ferrero's 7 million unique algorithm-designed Nutella labels** [9]: expensive, and Gulf families need to know the brand first.

REJECT – prices that change automatically: families could feel cheated, and Kazakhstan's 2026 amendments ban decisions made only by automated processing [5]. **Why overall:** Bayan Sulu is much smaller than Ferrero, with less customer data, so it should run one small Gulf pilot where AI prepares and a named person approves, then measure sales per store and repeat purchases before scaling.

E. Evidence

Sources (IEEE identifiers/URLs), screenshots or diagrams to include in Evidence Folder:

[1] L. J. Gitman et al., Introduction to Business, 2nd ed. Houston, TX, USA: OpenStax, 2026, chs. 11–13. [Online]. Available: https://openstax.org/details/books/introduction-business-2e

[2] JSC "Bayan Sulu," "About Us," bayansulu.kz. [Online]. Available: https://bayansulu.kz/en/about/o-kompanii/ (accessed Sep. 16, 2026).

[4] KazFoodProducts, "Bayan Sulu Confectionery Captivates Gulfood - 2026 and Expands into New Markets," press release, Jan. 30, 2026. [Online]. Available: https://kfp.kz/en/press-center/baian-sulu-konditerlik-fabrikasy-gulfood-2026-kormesin-bagyndyrdy (accessed Oct. 14, 2026).

[5] A. Shyngyssov and B. Kadyrov, "Republic of Kazakhstan Data Protection Law: Amendments in 2026," Morgan Lewis LawFlash, Jul. 14, 2026. [Online]. Available: https://www.morganlewis.com/pubs/2026/07/republic-of-kazakhstan-data-protection-law-amendments-in-2026

[7] National Institute of Standards and Technology, "Artificial Intelligence Risk Management Framework (AI RMF 1.0)," NIST AI 100-1, Jan. 2023, doi: 10.6028/NIST.AI.100-1.

[8] Google Cloud, "Ferrero Case Study." [Online]. Available: https://cloud.google.com/customers/ferrero (accessed Sep. 17, 2026).

[9] Futurism, "An Algorithm Designed 7 Million One-Of-A-Kind Labels for a Nutella Campaign." [Online]. Available: https://futurism.com/an-algorithm-designed-7-million-one-of-a-kind-labels-for-a-nutella-campaign (accessed Sep. 17, 2026).

[10] Alloy.ai, "Ferrero USA Selects Alloy.ai for Digital Supply Chain Management." [Online]. Available: https://alloy.ai/alloy-selected-by-ferrero-usa-for-digital-supply-chain-management (accessed Sep. 16, 2026).

Numbering follows the A3 photo essay reference list ([3] StockAnalysis and [6] Baker McKenzie are cited in other visuals).

Evidence Folder: Visual 4 – screenshot of the Gulfood press release [4]; Visual 3 – customer journey diagram (friction at Consideration); Visual 5 – people-only vs. with-AI comparison (method); Visual 6 – result; Visual 10 – transfer verdict; own shelf photo of Bayan Sulu products, Astana (Visual 1).

Part 2. Week 7 artifact: AI/data workflow and risk register

WEEK 7. AI/DATA WORKFLOW & RISK REGISTER

Student: Assel Saginayeva Group: AIB-2602 Company B: JSC Bayan Sulu (Company A: Ferrero)
Goal: show how information supports a business decision and how Company B governs the workflow. Use the Week 6 customer/marketing use case where possible.

A. Business decision and systems

Decision/use case; who benefits; what system(s) are involved (e.g., CRM, MIS, analytics):
Decision: for the Gulf pilot (one product line, one UAE channel), which launch ad versions go to which customer group and language, and how much stock to send to each shop/channel. This continues the Week 6 use case: launch messages for halal-seeking Gulf families that answer "Is it halal? Is it good?" [4].

Who benefits: Gulf families (clear messages in their own language, product on the shelf); the export manager and importer (right stock, less waste); the brand manager (faster ad drafts). **Systems:** TPS for orders, invoices and deliveries; a marketing data mart (the marketing slice of the data warehouse); AI used as a decision support system (DSS) for forecasts and ad drafts; MIS monthly reports [1].

Data vs information: what raw data enters, what useful information is produced?
Raw data in: order and delivery records from the TPS (by shop, date, SKU); loyalty sign-ups only where people agreed (language preference, emirate, contact); public review and competitor-price summaries with no personal data; brand facts and halal/label rules [2], [4].

Useful information out: a demand forecast per shop and channel; a short list of liked flavours and box sizes; ad options per group in Arabic, English and Kazakh; and an MIS statement a manager can act on, e.g. "waffle sales in one channel fell this month" [1] (photo essay, Visual 8).

B. Workflow map

Draw or insert your diagram here: DATA SOURCE → COLLECTION / ACCESS → PROCESSING / AI ANALYSIS → HUMAN VERIFICATION → BUSINESS DECISION → CUSTOMER ACTION → MONITORING

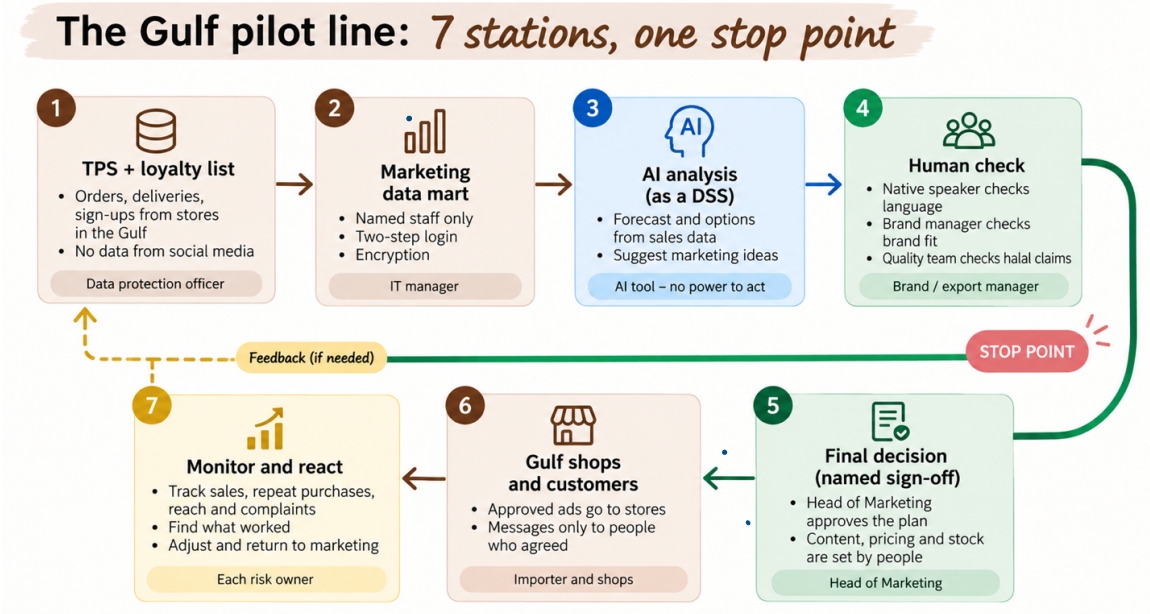


Figure: Gulf pilot line – 7 stations from data source to monitoring, with a stop point at the human check and a feedback line (photo essay, Visual 7; diagram drawn by the author in Canva).

DATA SOURCE (TPS orders, consented loyalty sign-ups) → **COLLECTION / ACCESS** (marketing data mart, named access) → **PROCESSING / AI ANALYSIS** (AI as DSS: forecast + ad drafts) → **HUMAN VERIFICATION** (native speaker, brand manager, export manager) → **BUSINESS DECISION** (Head of Marketing signs) → **CUSTOMER ACTION** (approved ad / stock reaches the UAE shop) → **MONITORING** (monthly checks, triggers, fallback).

For each stage: actor/system, inputs/outputs, permissions and data movement:
1 Data source – TPS + loyalty list. In: orders, deliveries, consented sign-ups. Out: raw records. Nothing is taken from social media [5].

2 Collection / access – marketing data mart (IT manager). Data moves from the TPS into the data mart; access by named staff only, two-step login, encryption [6].

3 Processing / AI analysis – AI tool as DSS. In: aggregated sales and brand facts – no names or contacts in prompts. Out: forecast and ad options. Permission: suggest only; it cannot publish, order or set prices.

4 Human verification – native speaker, brand manager, quality team, export manager. In: AI options. Out: corrected, approved options.

5 Business decision – Head of Marketing. Signs the final ad set and stock plan; every approval saved with name and date [5], [7].

6 Customer action – importer and UAE shops. Approved ads go out through the importer's channels; loyalty messages only to people who consented.

7 Monitoring – each risk owner. Monthly: forecast error, fair reach across groups and languages, complaints; results go back into planning.

Human review: who approves what, and using which criteria?
Native speaker + brand manager approve every ad: correct meaning, brand voice of a 50+ year brand, no health claims, no stereotypes. **Quality team** approves every halal/label claim: only claims backed by a valid certificate.

Export manager approves the stock plan: forecast compared with real distributor orders. **Head of Marketing** gives the final named sign-off; no name on a decision → the campaign pauses (photo essay, Visual 9).

Escalation for uncertain/high-risk cases; manual fallback for outages:
Escalation: uncertain or high-risk cases (a halal or health claim, a customer complaint, missing consent, a forecast far from real orders) go risk owner → Head of Marketing; AI never decides about a customer alone [5].

Manual fallback: if the AI tool or data mart is down, or forecast error is over 25% for 2 months, plan from distributor orders and an offline customer list, and use the Week 6 human-only route (agency + translator) for ads. Thresholds are the author's proposals, not company data.

C. Governance risk register — at least two substantive risks

Risk and affected data	Likelihood / impact	Control + owner	Monitoring / evidence
Privacy – customer data used without consent, or public posts reused for ads. Data: loyalty sign-ups (contact, language, emirate).	Likelihood: Medium. Impact: High – legal breach under 2026 rules [5] and loss of trust in a new market.	Use only consented sign-ups; never take data from social media; no personal data in AI prompts. **Owner: Data protection officer.**	Consent log; complaints register. Trigger: any complaint or missing consent → stop using that data, review within 48 h.
Reliability – wrong forecast, bad translation, or an ad with a false	Likelihood: High (new market, little sales history). Impact: High –	Forecast compared with real orders; native speaker + brand manager approve	Monthly forecast-error report; approval log. Trigger: error > 25% for 2

halal/health claim. Data: TPS sales, AI ad drafts.	wasted stock or a misleading claim.	every ad; quality team checks claims [7]. **Owner: Export manager.**	months → plan from distributor orders.
Security – one data mart holds the whole customer list, so one leak exposes everyone [1]. Data: data mart, loyalty list.	Likelihood: Low–Medium. Impact: High.	Encryption, named access, two-step login; breaches reported under 2026 rules [6]. **Owner: IT manager.**	Access log reviewed monthly; incident register; breach → notify and lock accounts.
Bias & accountability – some groups or languages get fewer offers; an AI output is used without anyone signing. Data: offer lists, ad sets.	Likelihood: Medium. Impact: Medium–High – unfair treatment; automated-only decisions are banned [5].	Monthly fairness check across groups and languages (Marketing manager); named sign-off on every decision. **Owner: Head of Marketing.**	Offers-per-group report; approval log with name and date. Trigger: no name on a decision → pause the campaign.

Consider privacy, security, bias, reliability, accountability, escalation and fallback. Do not list a risk without an actionable control.

D. Governance decision and transfer

Which risk matters most for Company B and why? What is the residual risk?

Privacy and consent matter most. Bayan Sulu is new in the Gulf and has little customer data; one misuse of personal data would damage trust before the brand is known, and Kazakhstan's 2026 amendments ban decisions made only by automated processing and tighten breach rules [5], [6].

Residual risk (medium-low): even with consent-only data and named access, the importer partner and staff still handle contact data, and AI drafts can still contain errors – so human review and monthly checks remain necessary.

Company A practice + verified source; transfer verdict and Kazakhstan-specific justification:

Ferrero practice: Ferrero built a consumer data platform on Google Cloud BigQuery ("Raw Data" by fifty-five) that splits its consumer database into real-time clusters for personalised content and media buying [8]. Ferrero USA also uses Alloy.ai for demand forecasting [10].

Verdict: ADAPT the BigQuery practice, PILOT forecasting (as in Week 6 and photo essay Visual 10). Kazakhstan-specific reasons: Bayan Sulu sells mainly through distributors and has far less online data than Ferrero; the 2026 personal-data amendments require consent-based processing, human involvement in decisions and breach reporting [5], [6]. So start with simple groups from distributor sales and consented sign-ups, with named human sign-off – not real-time automated personalisation.

Sources (IEEE; numbering follows the A3 photo essay and the Week 6 artifact):

[1] L. J. Gitman et al., Introduction to Business, 2nd ed. Houston, TX, USA: OpenStax, 2026, chs. 11–13. [Online]. Available: https://openstax.org/details/books/introduction-business-2e

[2] JSC "Bayan Sulu," "About Us," bayansulu.kz. [Online]. Available: https://bayansulu.kz/en/about/o-kompanii/ (accessed Sep. 16, 2026).

[4] KazFoodProducts, "Bayan Sulu Confectionery Captivates Gulfood - 2026 and Expands into New Markets," press release, Jan. 30, 2026. [Online]. Available: https://kfp.kz/en/press-center/baian-sulu-konditerlik-fabrikasy-gulfood-2026-kormesin-bagyndyrdy (accessed Oct. 14, 2026).

[5] A. Shyngyssov and B. Kadyrov, "Republic of Kazakhstan Data Protection Law: Amendments in 2026," Morgan Lewis LawFlash, Jul. 14, 2026. [Online]. Available: https://www.morganlewis.com/pubs/2026/07/republic-of-kazakhstan-data-protection-law-amendments-in-2026

[6] Baker McKenzie, "Kazakhstan: New Amendments to Personal Data Legislation," Jul. 21, 2026. [Online]. Available: https://www.bakermckenzie.com/en/insight/publications/2026/07/kazakhstan-new-amendments-to-personal-data-legislation

[7] National Institute of Standards and Technology, "Artificial Intelligence Risk Management Framework (AI RMF 1.0)," NIST AI 100-1, Jan. 2023, doi: 10.6028/NIST.AI.100-1.

[8] Google Cloud, “Ferrero Case Study.” [Online]. Available: https://cloud.google.com/customers/ferrero (accessed Sep. 17, 2026).

[10] Alloy.ai, “Ferrero USA Selects Alloy.ai for Digital Supply Chain Management.” [Online]. Available: https://alloy.ai/alloy-selected-by-ferrero-usa-for-digital-supply-chain-management (accessed Sep. 16, 2026).

Links to other work: Week 6 artifact (use case and comparison); photo essay Visuals 7 (workflow), 8 (data systems), 9 (risk controls) and 10 (transfer verdict).

Weekly Record 6: Marketing and Customer

Assel Saginayeva, AIB 2602

Chapters 11–12, Marketing and Customers · Company B: Bayan Sulu · Full record, Fields 1–6 · Develops into Capstone Section 5b, Marketing and customer.

Field 1 · Concept

Key terms from Chapters 11 and 12 [1]:

- Marketing concept: first find out what customers need, then make a product for that need and still make a profit.
- Customer value: what the customer gets for what they pay. This is not only money, but also time and effort.
- Customer satisfaction: the customer feels the product was as good as they expected, or better.
- Relationship marketing: trying to keep the same customers for years instead of selling to them once.
- Environmental scanning: watching what happens outside the company, like competitors, prices and new laws.
- Market segmentation: dividing customers into groups (by region, age, lifestyle or the benefit they want) and then choosing a target market.
- Marketing mix (4Ps): product, price, place and promotion. The book also adds people as a fifth P.
- Distribution channel: the way a product travels from the factory to the shop.

Concept Check: What is the difference between customer value and customer satisfaction?

I understand it like this. Value is about the moment before you buy, when you look at the price and decide if it is worth it. Satisfaction comes after, when you have eaten the chocolate and you either want it again or you don't. Good value can win the first sale, but people only come back if they are satisfied. I think AI can help a little with value, for example by testing prices. Satisfaction is something only real customers can tell you.

Field 2 · Application

For the practice task I compared two ways Bayan Sulu could do its marketing: only with people, or with AI helping. I used the 4Ps and the customer journey from the chapter.

I wanted a real situation, so I used Gulfood. In January 2026 Bayan Sulu went to this exhibition in Dubai. It was the company's first international exhibition, and they had B2B meetings with partners from the UAE [2]. Buyers in the Gulf look for halal food, so my target segment is halal-seeking families in the Gulf. It is a new market for Bayan Sulu, which makes it a good place to think about AI.

4P / step	Only people	With AI	Risk and who checks
Research	Talk to people at the	AI reads online reviews	AI does not know local

4P / step	Only people	With AI	Risk and who checks
	trade show, visit some shops, ask the distributor	and competitor prices in the UAE (Arabic and English)	taste, so a marketer still visits shops
Product	Send the same best-sellers we sell in Kazakhstan	AI looks at reviews to see which flavours and box sizes Gulf families like	Halal and label rules; the quality team approves every label
Price	Check competitor prices by hand	AI tests different prices and suggests a range	Changing prices can feel unfair; the export manager sets the final price
Place	Find one importer from the Gulfood contacts	AI predicts demand for supermarkets and online shops	Bad data means a bad forecast; compare it with real orders every month
Promotion	One agency makes one campaign	AI writes many ad versions in Arabic, English and Kazakh	Bad translations or ads that don't look like Bayan Sulu; a native speaker and the brand manager approve each ad
Customer journey (awareness, purchase, loyalty)	Same message for everybody	A different message for each group, and a loyalty offer after the first purchase	Customer data has to follow Kazakhstan's law; AI cannot decide about a customer alone [3]

Looking at the table, AI is clearly faster and cheaper for research, translation and making lots of ad versions. But people are still better at taste and trust, and they know how Bayan Sulu should sound. The brand is more than 50 years old [2], and I don't think speed is worth risking that. So I think AI should prepare the options and a person should choose. In every row there is someone who signs off at the end.

Field 3 · Verification and AI

What I checked:

- Gulfood facts. I checked the dates (26–30 January), the first Kazakhstan pavilion, the B2B meetings and the quote from the Head of Marketing on the KazFoodProducts press release [2]. Everything matched. One thing I changed: the article only talks about meetings, not signed contracts, so I call the Gulf a possible new market and not a current one.

- Data law in Kazakhstan. I read a law-firm summary from July 2026 [3]. Under the new rules, a decision cannot be made only by automated processing, and people can object to it. That is why the last row of my table says AI cannot decide about a customer alone.
- Ferrero's AI marketing (the 7 million Nutella labels and the customer groups in Google BigQuery). I already checked these in Weekly Record 3, so I took them from my Evidence Folder [4], [5] instead of searching again.
- The key terms, which I compared with the glossary in the OpenStax book [1].

AI use: I used Claude (Anthropic) to help me find the key terms and to make a first version of the table. After that I checked the facts and sources myself and rewrote the text.

Field 4 · Transfer

I think Bayan Sulu should use AI to prepare marketing work, but people should keep the brand voice and make every decision that touches the customer.

Choice	AI idea	My reason for Bayan Sulu
Adopt	AI that scans reviews and competitor prices for research	It is cheap and low risk, and it helps before entering the Gulf
Adapt	Customer groups, like Ferrero does with Google BigQuery [5]	Bayan Sulu has much less online data, so it can start with simple groups from marketplace and distributor sales, with consent
Pilot	AI ad versions in different languages	Try it on one product line and one UAE channel first, with a native speaker checking
Delay	Unique AI-designed labels, like Ferrero's 7 million Nutella jars [4]	Nice idea, but expensive. People in the Gulf need to know the brand first
Reject	Prices that change automatically	Families could feel cheated, and the 2026 law does not allow decisions made only by automated processing [3]

Field 5 · Evidence

This week I added two new sources to the Evidence Folder, [2] and [3]. Sources [4] and [5] are from Weekly Record 3.

[1] L. J. Gitman, C. McDaniel, A. Shah, M. Reece, L. Koffel, B. Talsma, and J. C. Hyatt, *Introduction to Business*, 2nd ed. Houston, TX, USA: OpenStax, 2026, chs. 11–12. [Online]. Available: https://openstax.org/details/books/introduction-business-2e

[2] KazFoodProducts, "Bayan Sulu Confectionery Captivates Gulfood - 2026 and Expands into New Markets," press release, Jan. 30, 2026. [Online]. Available: https://kfp.kz/en/press-center/baian-sulu-konditerlik-fabrikasy-gulfood-2026-kormesin-bagyndyrdy (accessed Oct. 14, 2026).

[3] A. Shyngyssov and B. Kadyrov, "Republic of Kazakhstan Data Protection Law: Amendments in 2026," Morgan Lewis LawFlash, Jul. 14, 2026. [Online]. Available: https://www.morganlewis.com/pubs/2026/07/republic-of-kazakhstan-data-protection-law-amendments-in-2026 (accessed Oct. 14, 2026).

[4] Futurism, "An Algorithm Designed 7 Million One-Of-A-Kind Labels for a Nutella Campaign," Futurism. [Online]. Available: https://futurism.com/an-algorithm-designed-7-million-one-of-a-kind-labels-for-a-nutella-campaign (accessed Sep. 17, 2026).

[5] Google Cloud, "Ferrero Case Study," Google Cloud. [Online]. Available: https://cloud.google.com/customers/ferrero (accessed Sep. 17, 2026).

Field 6 · Capstone link

This record will become Capstone Section 5b, Marketing and customer. The comparison table will be the main exhibit, because it shows where AI saves time and where a person has to approve.

In the Capstone I want to turn the Pilot row into a real test with costs: one product line, one channel in the Gulf, and ads that a native speaker approves. I would measure sales per store and how many customers buy again. It also connects to Section 6 (Week 7), because the customer data from marketing needs its own rules.

Weekly Record 7: Technology and Data

Assel Saginayeva, AIB2602

Chapter 13, Using Technology to Manage Information · Company B: Bayan Sulu · Reduced record, Fields 1–2 only (Fields 3–5 absorbed into A3 Photo Essay) · Develops into Capstone Section 6, Governance, verification and assurance.

Field 1 · Concept

Key terms from Chapter 13 [1]:

- Data and information: data are raw facts, like one sale at a till. Information is data that has been organised so a manager can use it, for example "waffle sales in Almaty went down this month".
- Management information system (MIS): the methods and equipment that give managers information about every part of the business.
- Transaction processing system (TPS): handles the daily work like orders, invoices and deliveries. The other systems get their raw data from it.
- Batch and online processing: batch updates data in groups, for example once a night. Online (real-time) updates it as soon as it comes in.
- Decision support system (DSS): lets managers try "what if" models before they decide.
- Expert system: uses AI to give advice, a bit like asking a human expert.
- Data warehouse and data mart: a warehouse puts all the company's databases in one place. A data mart is a smaller part of it for one area, like marketing.
- Data mining: looking through big data sets for hidden patterns that help predict what people will do.
- Knowledge management: collecting and sharing what the company knows, so it doesn't disappear when people leave.
- Data security: protecting data from viruses, hackers, theft and simple mistakes, with things like firewalls, encryption, passwords, backups and staff training.

Concept Check: Why are privacy advocates worried about data warehouses and data mining?

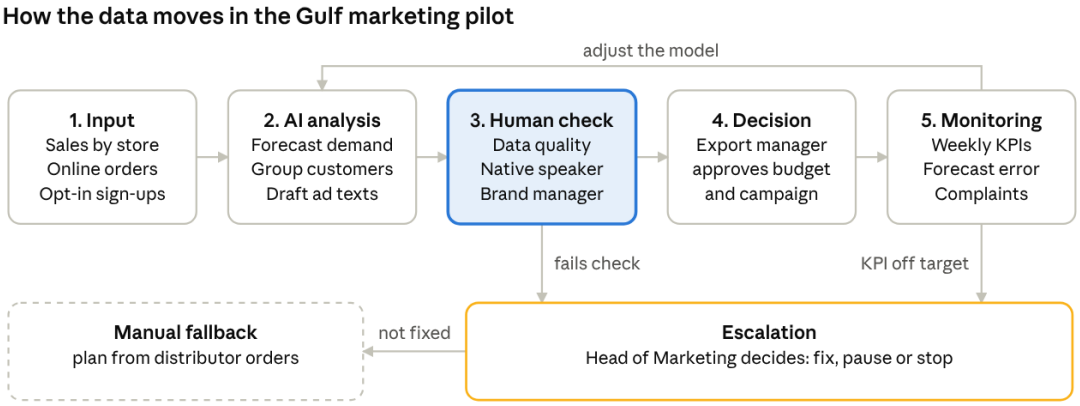
When a company joins a lot of databases together, it can find out much more about one person than that person ever agreed to share. Then data mining can guess things about them, like their income or health, and the company might treat them differently because of it. Most customers have no idea this is happening. In my opinion, the more a company can do with data, the more rules it needs about what it is allowed to do.

Field 2 · Application

For the practice task I made an AI/data workflow and a risk register for Bayan Sulu's marketing pilot in the Gulf.

It continues my Week 6 record. There I suggested a small AI marketing test for halal-seeking families in the Gulf. This week I wanted to show how the data actually moves, where people check it, and what

happens if something goes wrong. Using the chapter's words, the TPS collects the sales data, a marketing data mart keeps it, and the AI works like a decision support system. It gives options, but managers make the decision.



Gulf marketing pilot · data workflow, 5 steps, escalation and fallback

The AI never acts by itself. At step 3 a person can stop anything it makes, and at step 5 problems go up to escalation or to the manual plan.

Risk register. I gave every risk one owner, a trigger for when to escalate, and a manual backup. The numbers are my own suggestions for the pilot, not real company data.

Risk	What could go wrong	How we control it	Owner	Escalation trigger and backup
Privacy	Customer data is used without consent, or data from public posts is reused for ads	Only use loyalty sign-ups where people agreed; don't take data from social media [2]	Data protection officer	Any complaint or missing consent: stop using that data and review it within 48 hours
Security	The customer list gets leaked or hacked	Encryption, only named staff have access, two-step login, monthly backup [1], [3]	IT manager	Any breach: report it under the 2026 breach register rules [3] and switch to an offline list
Bias	AI shows offers only to richer areas or to one language group	Check every month that all groups and languages get a fair share	Marketing manager	A group gets less than half its share: fix it by hand

Risk	What could go wrong	How we control it	Owner	Escalation trigger and backup
Reliability	A wrong forecast, a bad translation, or an ad with a false claim (for example about health)	Compare the forecast with real orders; a native speaker and the brand manager approve every ad	Export manager	Forecast error over 25% for 2 months: go back to planning from distributor orders
Accountability	Nobody knows who approved a campaign or a price	Every approval is saved with a name and date; AI never publishes ads or sets prices by itself [2], [4]	Head of Marketing	A decision with no name on it: pause the campaign until someone signs off

I built the register around two rules. Kazakhstan's 2026 law does not allow decisions made only by automated processing [2], and the NIST framework says every AI system needs clear human roles [4]. So in this workflow the AI can only suggest. A named person has to act.

Sources

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[2] A. Shyngyssov and B. Kadyrov, "Republic of Kazakhstan Data Protection Law: Amendments in 2026," Morgan Lewis LawFlash, Jul. 14, 2026. [Online]. Available: https://www.morganlewis.com/pubs/2026/07/republic-of-kazakhstan-data-protection-law-amendments-in-2026

[3] Baker McKenzie, "Kazakhstan: New Amendments to Personal Data Legislation," Jul. 21, 2026. [Online]. Available: https://www.bakermckenzie.com/en/insight/publications/2026/07/kazakhstan-new-amendments-to-personal-data-legislation

[4] National Institute of Standards and Technology, "Artificial Intelligence Risk Management Framework (AI RMF 1.0)," NIST AI 100-1, Jan. 2023, doi: 10.6028/NIST.AI.100-1.